

MB 402: Operations Research

Course Code: MB 402

Course Name: Operations Research

L-T-P-C: 3-0-0-3

Prerequisites: None

Intended for: IMBA

Mutual Exclusion: MA 515

Approval: 59th BoA (I-MBA curriculum); course proposal approved in the 65th BoA

Preamble

Operations Research (OR) provides scientific and quantitative techniques for managerial decision-making and optimal utilization of organizational resources. The course introduces management students to analytical models and optimization methods widely used in production, logistics, finance, marketing, supply chain, and service industries. Emphasis is placed on problem formulation, interpretation of solutions, and managerial applications through practical business cases. The course integrates theoretical concepts with computer-based laboratory exercises using optimization software packages. Students will develop analytical, computational, and decision-making skills required for solving real-life managerial problems in competitive and dynamic business environments.

Course Modules

- **Introduction to Operations Research and Model Building:** Nature, scope, objectives, and significance of Operations Research; characteristics and phases of OR; applications of OR in management; classification of models; model formulation and limitations of OR techniques in business decision-making. (5 Hours)
- **Linear Programming Problem and Solution Techniques:** Formulation of linear programming problems from business situations; graphical solution method; simplex method; Big-M method; two-phase method; degeneracy, infeasibility, and unbounded solutions; managerial applications of linear programming in production, transportation, marketing, and finance. (10 Hours)
- **Transportation and Assignment Problems:** Structure and formulation of transportation problems; methods for obtaining initial feasible solutions including North-West Corner Method, Least Cost Method, and Vogel's Approximation Method; optimality test using MODI method; assignment problem formulation and Hungarian method; practical applications in scheduling and resource allocation. (7 Hours)
- **Network Analysis and Project Management:** Construction of project networks; CPM and PERT techniques; identification of critical path; float analysis; project scheduling; project crashing and time-cost optimization; managerial applications in project planning and control. (6 Hours)
- **Inventory Control Models:** Concepts of inventory and inventory costs; deterministic inventory models; Economic Order Quantity (EOQ); quantity discount models; inventory decision-making in manufacturing and retail systems. (4 Hours)
- **Queueing Theory and Simulation:** Characteristics of queueing systems; single-channel and multi-channel queue models; performance measures of waiting lines; applications in banking, healthcare, and service industries; introduction to simulation and Monte Carlo techniques. (4 Hours)

- **Decision Theory and Game Theory:** Decision-making under certainty, risk, and uncertainty; payoff matrix; decision criteria; expected monetary value; decision tree analysis; game theory concepts; two-person zero-sum games; pure and mixed strategies; managerial applications of competitive decision-making. (6 Hours)

Laboratory/Practical/Tutorial Modules

NA

Text Books

1. Anderson D.R., Sweeney D.J., Williams T.A., and Martin K., *Quantitative Methods for Business*, 13th Edition, Cengage Learning, USA, 2019.
2. Hiller F.S. and Lieberman G.J., *Introduction to Operations Research*, 11th Edition, McGraw-Hill Education, USA, 2020.

References

1. Winston W.L., *Operations Research: Applications and Algorithms*, 4th Edition, Cengage Learning, USA, 2014.
2. Hamdy A. Taha, *Operations Research: An Introduction*, 10th Edition, Pearson Education, USA, 2017.
3. Render B., Stair R.M., Hanna M.E., and Hale T.S., *Quantitative Analysis for Management*, 14th Edition, Pearson Education, USA, 2020.